Lab 2 practice: exact equations and substitution (Zill §2.4 and §2.5)

Prep for the Lab 2 quiz. Closed book, about 10 minutes per problem. Eight problems, one per method, with the last one solvable two ways. Classify first, then solve, then check by differentiating the answer. Every solution was checked with `lab2_check.py`.

Exact. Write the equation as $M\,dx + N\,dy = 0$. It is exact when $M_y = N_x$. Then $F = \int M\,dx + g(y)$, and g comes from $F_y = N$. The answer is $F(x, y) = c$.

Not exact: integrating factor. Multiply through by μ and test again.

- If $\dfrac{M_y - N_x}{N}$ depends on x only: $\mu(x) = \exp\Big(\displaystyle\int \frac{M_y - N_x}{N}\,dx\Big)$
- If $\dfrac{N_x - M_y}{M}$ depends on y only: $\mu(y) = \exp\Big(\displaystyle\int \frac{N_x - M_y}{M}\,dy\Big)$

Check whether μ lost a solution (a line where μ is undefined or zero).

Substitutions.

Form	Substitution	Result
homogeneous: $y' = f(y/x)$, or M, N of equal degree	$y = ux, \quad y' = u + xu'$	separable in u, x
Bernoulli: $y' + P(x)y = f(x)y^n,\ n \neq 0, 1$	$u = y^{1-n}$	linear: $u' + (1-n)Pu = (1-n)f$
$y' = f(Ax + By + C),\ B \neq 0$	$u = Ax + By + C, \quad u' = A + By'$	separable: $u' = A + Bf(u)$

Go back to y at the end. State where the answer is valid.

Problem 1

$(2xy^2 + e^x)\,dx + (2x^2y - 3y^2)\,dy = 0$

Solution: plain exact

Differential form with $M = 2xy^2 + e^x$ and $N = 2x^2y - 3y^2$. Test:

$$M_y = 4xy, \qquad N_x = 4xy$$

They match, so the equation is exact and a function F exists with $F_x = M$ and $F_y = N$.

Integrate M with respect to x (treat y as a constant) and add an unknown function of y:

$$F = \int (2xy^2 + e^x)\, dx = x^2y^2 + e^x + g(y)$$

Differentiate in y and match N:

$$F_y = 2x^2y + g'(y) = 2x^2y - 3y^2 \quad \Longrightarrow \quad g'(y) = -3y^2, \quad g(y) = -y^3$$

$$\boxed{x^2y^2 + e^x - y^3 = c}$$

Check: $F_x = 2xy^2 + e^x = M$ and $F_y = 2x^2y - 3y^2 = N$. The solution is implicit and valid wherever it defines y as a differentiable function of x, which fails where $N = 0$.

Problem 2

$\left(\frac{y}{x}+6x\right)dx+(\ln x-2)\,dy=0, \qquad y(1)=5$

Solution: exact initial value problem

Here $M = y/x + 6x$ and $N = \ln x - 2$, defined for $x > 0$. Test:

$$M_y = \frac{1}{x}, \qquad N_x = \frac{1}{x}$$

Exact. Integrate M in x:

$$F = \int \left(\frac{y}{x} + 6x\right) dx = y \ln x + 3x^2 + g(y)$$

$$F_y = \ln x + g'(y) = \ln x - 2 \quad \Longrightarrow \quad g(y) = -2y$$

So the general solution is $y \ln x + 3x^2 - 2y = c$. Apply $y(1) = 5$ to the whole expression:

$$5(0) + 3 - 10 = c \quad \Longrightarrow \quad c = -7$$

Solve for y, which is possible here because y appears linearly:

$$y(\ln x - 2) = -7 - 3x^2$$

$$\boxed{y = \frac{3x^2 + 7}{2 - \ln x}, \qquad 0 < x < e^2}$$

Check: $y(1) = 10/2 = 5$. The interval is the one that contains $x = 1$: it needs $x > 0$ for $\ln x$ and $1/x$, and $2 - \ln x \neq 0$ rules out $x = e^2$.

Problem 3

$y\,dx + (x^2y - x)\,dy = 0$

Solution: integrating factor in x

$M = y$ and $N = x^2y - x$.

$$M_y = 1, \qquad N_x = 2xy - 1$$

Not exact. Try the x test first:

$$\frac{M_y - N_x}{N} = \frac{2 - 2xy}{x^2y - x} = \frac{-2(xy-1)}{x(xy-1)} = -\frac{2}{x}$$

That depends on x only. (The y test gives $(N_x - M_y)/M = (2xy - 2)/y$, which still has an x in it, so it fails.)

$$\mu(x) = \exp\left(-2\int \frac{dx}{x}\right) = x^{-2}$$

Multiply through:

$$\frac{y}{x^2}\,dx + \left(y - \frac{1}{x}\right)dy = 0$$

Recheck: $M_y = 1/x^2$ and $N_x = 1/x^2$. Exact now. Integrate M in x:

$$F = -\frac{y}{x} + g(y), \qquad F_y = -\frac{1}{x} + g'(y) = y - \frac{1}{x} \quad \Longrightarrow \quad g(y) = \frac{y^2}{2}$$

$$\boxed{\frac{y^2}{2} - \frac{y}{x} = c, \qquad x \neq 0}$$

Check: $F_x = y/x^2 = \mu M$ and $F_y = y - 1/x = \mu N$.

Lost solution: $x = 0$ satisfies the original equation ($dx = 0$ and $x^2y - x = 0$ there) but $\mu = x^{-2}$ is undefined on it. The line $y = 0$ is a solution too and is contained in the family at $c = 0$.

Problem 4

$y\,dx + (2x - ye^y)\,dy = 0$

Solution: integrating factor in y

$M = y$ and $N = 2x - ye^y$.

$$M_y = 1, \qquad N_x = 2$$

Not exact. The x test gives $(M_y - N_x)/N = -1/(2x - ye^y)$, which depends on both variables, so it fails. The y test:

$$\frac{N_x - M_y}{M} = \frac{2-1}{y} = \frac{1}{y}$$

$$\mu(y) = \exp\left(\int \frac{dy}{y}\right) = y$$

Multiply through:

$$y^2\,dx + (2xy - y^2e^y)\,dy = 0$$

Recheck: $M_y = 2y$ and $N_x = 2y$. Exact. Integrate M in x:

$$F = xy^2 + g(y), \qquad F_y = 2xy + g'(y) = 2xy - y^2e^y \quad \Longrightarrow \quad g'(y) = -y^2e^y$$

Two integrations by parts give $\int y^2e^y\,dy = e^y(y^2 - 2y + 2)$, so

$$\boxed{xy^2 - e^y(y^2 - 2y + 2) = c}$$

Check: $F_x = y^2 = \mu M$ and $F_y = 2xy - e^y(y^2 - 2y + 2) - e^y(2y - 2) = 2xy - y^2e^y = \mu N$.

The factor $\mu = y$ vanishes on $y = 0$. That line still satisfies the original equation ($dy = 0$ and $M = 0$), so nothing was lost.

Problem 5

$x\dfrac{dy}{dx} = y + xe^{y/x}, \qquad x > 0$

Solution: homogeneous

Divide by x: $\dfrac{dy}{dx} = \dfrac{y}{x} + e^{y/x}$. The right side depends on y/x only, so the equation is homogeneous. Let $y = ux$, so $y' = u + xu'$:

$$u + xu' = u + e^u \quad \Longrightarrow \quad xu' = e^u$$

Separate and integrate:

$$e^{-u}\,du = \frac{dx}{x} \quad \Longrightarrow \quad -e^{-u} = \ln x + c$$

Write $e^{-u} = C - \ln x$ with $C = -c$. Then $u = -\ln(C - \ln x)$ and $y = ux$:

$$\boxed{y = -x\ln(C - \ln x), \qquad 0 < x < e^C}$$

Check: $y' = -\ln(C - \ln x) + \dfrac{1}{C - \ln x}$. On the other side, $\dfrac{y}{x} + e^{y/x} = -\ln(C - \ln x) + \dfrac{1}{C - \ln x}$. They agree. The interval comes from needing $C - \ln x > 0$. Since e^u is never zero, no constant solution was divided out.

Problem 6

$x\dfrac{dy}{dx} + y = x^2y^3, \qquad y(1) = \dfrac{1}{2}$

Solution: Bernoulli

Divide by x to reach standard form:

$$y' + \frac{1}{x}\, y = x\, y^3$$

This is Bernoulli with $P = 1/x$, $f = x$ and $n = 3$. Use $u = y^{1-n} = y^{-2}$, so $u' = -2y^{-3}y'$. Divide the equation by y^3 and substitute:

$$y^{-3}y' + \frac{1}{x}y^{-2} = x \quad \Longrightarrow \quad -\frac{1}{2}u' + \frac{1}{x}u = x$$

Multiply by -2 to get the linear form $u' + (1-n)Pu = (1-n)f$:

$$u' - \frac{2}{x}\, u = -2x$$

Integrating factor: $\exp(-2\int dx/x) = x^{-2}$. Then

$$\frac{d}{dx}(x^{-2}u) = -\frac{2}{x} \quad \Longrightarrow \quad x^{-2}u = -2\ln x + c \quad \Longrightarrow \quad u = x^2(c - 2\ln x)$$

Return to y: $y^{-2} = x^2(c - 2\ln x)$. Apply $y(1) = \frac{1}{2}$, so $u(1) = 4$ and $c = 4$. The initial value is positive, so take the positive root:

$$\boxed{y = \frac{1}{x\sqrt{4 - 2\ln x}}, \qquad 0 < x < e^2}$$

Check: $y(1) = 1/\sqrt{4} = 1/2$. The interval needs $x > 0$ and $4 - 2\ln x > 0$. Dividing by y^3 lost the solution $y = 0$; it is not this IVP's solution.

Problem 7

$\frac{dy}{dx} = \tan^2(x + y)$

Solution: substitution $u = Ax + By + C$

The right side is a function of $x + y$ alone, so let $u = x + y$. Then $u' = 1 + y'$:

$$u' = 1 + \tan^2 u = \sec^2 u$$

Separable:

$$\cos^2 u \, du = dx \quad \Longrightarrow \quad \frac{u}{2} + \frac{\sin 2u}{4} = x + c$$

Multiply by 4 and put $u = x + y$ back:

$$2(x + y) + \sin\left(2(x + y)\right) = 4x + c_1$$

$$\boxed{2y - 2x + \sin(2x + 2y) = c_1}$$

Check by differentiating implicitly, with $F = 2y - 2x + \sin(2x + 2y)$:

$$F_x = -2 + 2\cos(2x + 2y), \qquad F_y = 2 + 2\cos(2x + 2y)$$

$$y' = -\frac{F_x}{F_y} = \frac{1 - \cos(2x + 2y)}{1 + \cos(2x + 2y)} = \tan^2(x + y)$$

using $1 - \cos 2\theta = 2\sin^2 \theta$ and $1 + \cos 2\theta = 2\cos^2 \theta$.

Problem 8

$(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0$

Solution: two methods, same answer

$M = 3xy + y^2$ and $N = x^2 + xy$. Test for exactness first:

$$M_y = 3x + 2y, \qquad N_x = 2x + y$$

Not exact. But M and N are both homogeneous of degree 2, and the x test also works. Two routes.

Method A: integrating factor in x.

$$\frac{M_y - N_x}{N} = \frac{x + y}{x(x + y)} = \frac{1}{x} \quad \Longrightarrow \quad \mu = \exp \int \frac{dx}{x} = x$$

Multiply through: $(3x^2y + xy^2)\,dx + (x^3 + x^2y)\,dy = 0$. Now $M_y = 3x^2 + 2xy = N_x$.

$$F = x^3y + \frac{x^2y^2}{2} + g(y), \qquad F_y = x^3 + x^2y + g'(y) = x^3 + x^2y \quad \Longrightarrow \quad g' = 0$$

So $x^3y + \frac{1}{2}x^2y^2 = c$, or equivalently $x^2y^2 + 2x^3y = C$.

Method B: homogeneous substitution. With $y = ux$ and $dy/dx = -M/N$, divide numerator and denominator by x^2:

$$u + xu' = -\frac{3u + u^2}{1 + u} \quad \Longrightarrow \quad xu' = -\frac{3u + u^2 + u + u^2}{1 + u} = -\frac{2u(u + 2)}{1 + u}$$

Separate and split with partial fractions, $\dfrac{1 + u}{u(u + 2)} = \dfrac{1}{2}\left(\dfrac{1}{u} + \dfrac{1}{u + 2}\right)$:

$$\frac{1}{2}\left(\frac{1}{u} + \frac{1}{u + 2}\right) du = -\frac{2\,dx}{x} \quad \Longrightarrow \quad \frac{1}{2}\ln|u(u + 2)| = -2\ln|x| + c$$

Exponentiate to get $u(u + 2)\,x^4 = C$, then use $u = y/x$:

$$\frac{y}{x}\left(\frac{y}{x} + 2\right)x^4 = x^2y(y + 2x) = C$$

$$\boxed{x^2y^2 + 2x^3y = C}$$

Both routes give the same family. Method A is shorter because the integration is trivial; method B needs the partial fractions but needs no test to find μ. The line $y = 0$ and the line $y = -2x$ (where $u = 0$ and $u = -2$ were divided out) are solutions and sit inside the family at $C = 0$, since $x^2y(y + 2x) = 0$.

Check: $F = x^2y^2 + 2x^3y$ has $F_x = 2xy^2 + 6x^2y = 2x\,(y^2 + 3xy)$ and $F_y = 2x^2y + 2x^3 = 2x\,(x^2 + xy)$, which are $2x\,M$ and $2x\,N$. So $F_x\,dx + F_y\,dy = 2x\,(M\,dx + N\,dy) = 0$.